

FC7xxx ETHTRCV Integration Manual

Rev.A0

Revision History

Revision	Date	Changes
A0	2025/1/22	Add support for FC7300F8MDQ

Table of Contents

Chapter 1 Introduction	4
1.1 Introduction	4
Chapter 2 Building	5
2.1 Dependencies on Other Modules	5
2.2 Files Required for Compile	5
2.3 Add Plug-ins	6
Chapter 3 Memory	7
3.1 Sections in Memory Map	7
Chapter 4 Exclusive Area	9
Chapter 5 Interrupt Service Routine (ISR).....	10
Chapter 6 Error Report	11
6.1 Det.....	11
6.2 Dem.....	12
Chapter 7 Function Calls.....	13
7.1 Function Calls during Startup	13
7.2 Function Calls during Shutdown.....	13
7.3 Function Calls during Wake-up.....	13
7.4 Function Calls during Runtime	13
Chapter 8 Other Requirements.....	14
8.1 Notification, Callback, Callout	14
8.2 Macros.....	14
Chapter 9 Integration steps	15

Chapter 1 Introduction

1.1 Introduction

This integration manual describes the integration requirements for EthTrcv module.

Chapter 2 Building

2.1 Dependencies on Other Modules

Module configuration dependency

- Mcu: This module provides the clock reference point for EthTrcv module.
- Common: This module is the basic module which used to choose the chip.
- Port: This module provides the IO multiplexing for EthTrcv module.
- ECUC: This module provides the ECUC partition informations for EthTrcv module.

Module build dependency

- Common: This module provides common type for other modules.
- Rte: This module provides APIs to protect/unprotect some parts of code from interrupts (Exclusive Areas).
- Det: This module is necessary for enabling Development error detection.
- Dem: This module is necessary for reporting diagnostic errors.

Module initialization dependency

- Mcu: To use the EthTrcv module, the user needs to initialize MCU module first.

2.2 Files Required for Compile

EthTrcv module files:

- MCAL/Src/EthTrcv/src/EthTrcv.c
- MCAL/Src/EthTrcv/src/EthTrcv_Hal.c
- MCAL/Src/EthTrcv/src/EthTrcv_Hw.c
- MCAL/Src/EthTrcv/include/EthTrcv.h
- MCAL/Src/EthTrcv/include/EthTrcv_Hal.h
- MCAL/Src/EthTrcv/include/EthTrcv_Types.h
- MCAL/Src/EthTrcv/include/EthTrcv_Hw.h
- MCAL/Src/EthTrcv/include/EthTrcv_Reg.h
- MCAL/Src/EthTrcv/include/EthTrcv_MemMap.h
- MCAL/Src/EthTrcv/include/EthTrcv_Version.h
- _MCAL_multicore_generate_8M/src/EthTrcv_PBcfg.c
- _MCAL_multicore_generate_8M/include/EthTrcv_Cfg.h

Common module files:

- MCAL/Src/Base/include/Mcal.h
- MCAL/Src/Base/include/Std_Types.h
- MCAL/Src/Base/include/Platform_Types.h
- MCAL/Src/Base/include/Compiler.h
- MCAL/Src/Base/include/Compiler_Cfg.h
- MCAL/Src/Base/include/CompilerDefinition.h
- MCAL/Src/Base/include/Eth_GeneralTypes.h
- MCAL/Src/Common/include/Scm_Reg.h

Det module files:

- Det.h
- Det.c

Dem module files:

- Dem.h
- Dem.c

Rte files:

- SchM_EthTrcv.h
- SchM_EthTrcv.c

2.3 Add Plug-ins

EthTrcv module plug-ins are developed for EB tresos Studio, so, to use EthTrcv plug-ins on the EB tresos Studio, the user needs to add the EthTrcv module to the EB plug-ins folder first.

- 1) Copy the EthTrcv module(MCAL/EB_Plugins/eclipse/plugins/ EthTrcv) folder to EB tresos plug-ins (EB/tresos/plugins/) folder.
- 2) Set the EthTrcv module output location folder for the generated source file and header files.
- 3) Use EB tresos to configure the EthTrcv module and generate source and header files.

Chapter 3 Memory

3.1 Sections in Memory Map

Section Name	Section Type	Description
ETHTRCV_START_SEC_CONFIG_DATA_8 ETHTRCV_STOP_SEC_CONFIG_DATA_8 ETHTRCV_START_SEC_CONFIG_DATA_16 ETHTRCV_STOP_SEC_CONFIG_DATA_16 ETHTRCV_START_SEC_CONFIG_DATA_32 ETHTRCV_STOP_SEC_CONFIG_DATA_32 ETHTRCV_START_SEC_CONFIG_DATA_UNSPECIFIED ETHTRCV_STOP_SEC_CONFIG_DATA_UNSPECIFIED	Configuration Data	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (data).
ETHTRCV_START_SEC_CONST_BOOLEAN ETHTRCV_STOP_SEC_CONST_BOOLEAN ETHTRCV_START_SEC_CONST_8 ETHTRCV_STOP_SEC_CONST_8 ETHTRCV_START_SEC_CONST_16 ETHTRCV_STOP_SEC_CONST_16 ETHTRCV_START_SEC_CONST_32 ETHTRCV_STOP_SEC_CONST_32 ETHTRCV_START_SEC_CONST_UNSPECIFIED ETHTRCV_STOP_SEC_CONST_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read only (rodata).
ETHTRCV_START_SEC_CODE ETHTRCV_STOP_SEC_CODE ETHTRCV_START_SEC_RAMCODE ETHTRCV_STOP_SEC_RAMCODE ETHTRCV_START_SEC_CODE_AC ETHTRCV_STOP_SEC_CODE_AC	Code	Start and stop of memory Section for Code (text).
ETHTRCV_START_SEC_VAR_NO_INIT_BOOLEAN ETHTRCV_STOP_SEC_VAR_NO_INIT_BOOLEAN ETHTRCV_START_SEC_VAR_NO_INIT_8 ETHTRCV_STOP_SEC_VAR_NO_INIT_8 ETHTRCV_START_SEC_VAR_NO_INIT_16 ETHTRCV_STOP_SEC_VAR_NO_INIT_16 ETHTRCV_START_SEC_VAR_NO_INIT_32 ETHTRCV_STOP_SEC_VAR_NO_INIT_32 ETHTRCV_START_SEC_VAR_NO_INIT_UNSPECIFIED ETHTRCV_STOP_SEC_VAR_NO_INIT_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are never cleared and never initialized by startup code (bss).
ETHTRCV_START_SEC_VAR_INIT_BOOLEAN ETHTRCV_STOP_SEC_VAR_INIT_BOOLEAN ETHTRCV_START_SEC_VAR_INIT_8 ETHTRCV_STOP_SEC_VAR_INIT_8 ETHTRCV_START_SEC_VAR_INIT_16 ETHTRCV_STOP_SEC_VAR_INIT_16 ETHTRCV_START_SEC_VAR_INIT_32 ETHTRCV_STOP_SEC_VAR_INIT_32	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (data).

ETHTRCV_START_SEC_VAR_INIT_UNSPECIFIED ETHTRCV_STOP_SEC_VAR_INIT_UNSPECIFIED		
ETHTRCV_START_SEC_VAR_NO_INIT_BOOLEAN_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_NO_INIT_BOOLEAN_NO_CACHEABLE ETHTRCV_START_SEC_VAR_NO_INIT_8_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_NO_INIT_8_NO_CACHEABLE ETHTRCV_START_SEC_VAR_NO_INIT_16_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_NO_INIT_16_NO_CACHEABLE ETHTRCV_START_SEC_VAR_NO_INIT_32_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_NO_INIT_32_NO_CACHEABLE ETHTRCV_START_SEC_VAR_NO_INIT_UNSPECIFIED_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_NO_INIT_UNSPECIFIED_NO_CACHEABLE	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit, and be placed in noncacheable section. These variables are never initialized by startup code.
ETHTRCV_START_SEC_VAR_INIT_BOOLEAN_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_INIT_BOOLEAN_NO_CACHEABLE ETHTRCV_START_SEC_VAR_INIT_8_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_INIT_8_NO_CACHEABLE ETHTRCV_START_SEC_VAR_INIT_16_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_INIT_16_NO_CACHEABLE ETHTRCV_START_SEC_VAR_INIT_32_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_INIT_32_NO_CACHEABLE ETHTRCV_START_SEC_VAR_INIT_UNSPECIFIED_NO_CACHEABLE ETHTRCV_STOP_SEC_VAR_INIT_UNSPECIFIED_NO_CACHEABLE	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit, and be placed in noncacheable section. These variables are initialized by startup code.

Chapter 4 Exclusive Area

EthTrcv module using the services of Schedule Manger (SchM) for entering and exiting critical regions.

The following critical regions are used in the EthTrcv driver:

No critical region is used in the transceiver driver, because the MDIO clock is slow and the MDIO bus access is inefficient.

Users need to ensure that an Ethernet controller is used by only one transceiver, otherwise, appropriate critical regions are needed on upper layer.

Chapter 5 Interrupt Service Routine (ISR)

EthTrcv has no interrupt service routine.

Chapter 6 Error Report

6.1 Det

Function Name	Error Type
EthTrcv_Init	ETHTRCV_E_PARAM_POINTER
EthTrcv_SetTransceiverMode	ETHTRCV_E_UNINIT; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetTransceiverMode	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_SetTransceiverWakeupMode	ETHTRCV_E_UNINIT; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetTransceiverWakeupMode	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetBusWuReason	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_CheckWakeup	ETHTRCV_E_UNINIT; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_StartAutoNegotiation	ETHTRCV_E_UNINIT; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_TransceiverLinkStateRequest	ETHTRCV_E_UNINIT; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetLinkState	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetBaudRate	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetDuplexMode	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_SetPhyTestMode	ETHTRCV_E_UNINIT; ETHTRCV_E_INV_CTRL_IDX;

	ETHTRCV_E_INV_PARTITION
EthTrcv_SetPhyLoopbackMode	ETHTRCV_E_UNINIT; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetPhySignalQuality	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_SetPhyTxMode	ETHTRCV_E_UNINIT; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_RunCableDiagnostic	ETHTRCV_E_UNINIT; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetCableDiagnosticsResult	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetPhyIdentifier	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetMacMethod	ETHTRCV_E_UNINIT; ETHTRCV_E_PARAM_POINTER; ETHTRCV_E_INV_CTRL_IDX; ETHTRCV_E_INV_PARTITION
EthTrcv_GetVersionInfo	ETHTRCV_E_PARAM_POINTER
EthTrcv_MainFunction	ETHTRCV_E_UNINIT;

6.2 Dem

Function Name	Error Type
EthTrcv_Init	ETH_E_ACCESS
EthTrcv_SetTransceiverMode	ETH_E_ACCESS

Chapter 7 Function Calls

7.1 Function Calls during Startup

The API need be called is `void EthTrcv_Init(const EthTrcv_ConfigType* CfgPtr)`.

7.2 Function Calls during Shutdown

None

7.3 Function Calls during Wake-up

None

7.4 Function Calls during Runtime

None

Chapter 8 Other Requirements

8.1 Notification, Callback, Callout

- Notification

EthIf_TrcvModeIndication shall be called when the transceiver mode changed.

EthIf_SleepIndication shall be called by the corresponding EthTrcv, when a sleep indication was detected on the network.

- Callback

EthTrcv_ReadMiiIndication

EthTrcv_WriteMiiIndication

- Callout

None

8.2 Macros

Please have a look in various definitions available in Common module's include file Mcal.h for details.

- If OS is not used:

AUTOSAR_OS_NOT_USED need defined.

- If OS is used:

Do not define AUTOSAR_OS_NOT_USED.

Chapter 9 Integration steps

- 1) Configure EthTrcv module and generate configuration files (please refer to Building chapter for details).
- 2) Configure appropriate memory sections in linker file or other (please refer to Memory chapter for details).
- 3) Map interrupt notification to their vector locations (please refer to chapter ISR for details).
- 4) Build the EthTrcv module with dependent modules.